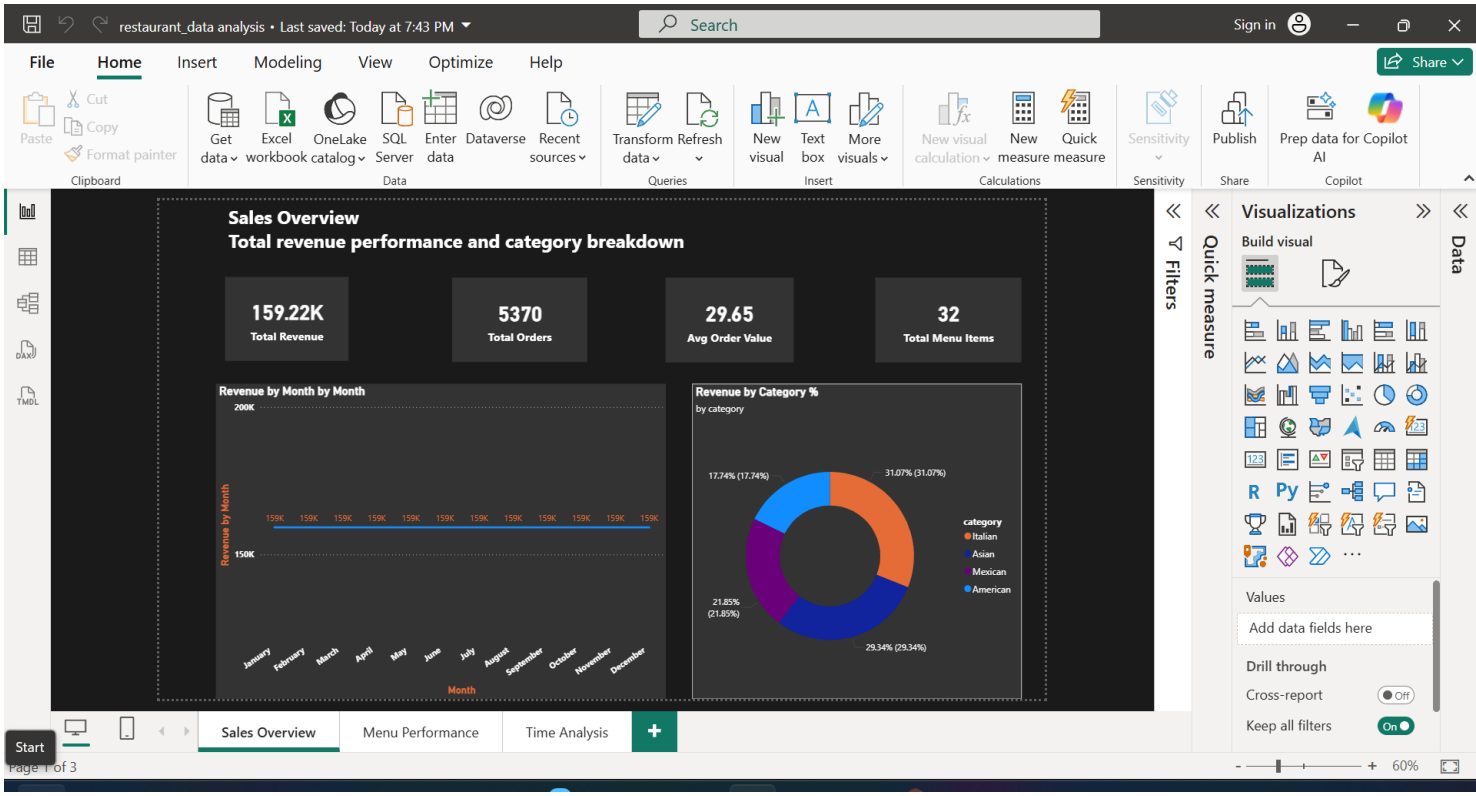
RESTAURANT ANALYTICS Power BI Project Report Data Cleaning & Visualization — 2026

Tools Used: Power Query | DAX Measures | Data Model | Power BI Visuals



Prepared in Power BI Desktop | Data: Excel Tables | 2026 Full Year

SECTION 1 — DATA MODEL & TABLES IN POWER BI

1.1 Data View — menu_items Table

The menu_items table was loaded into Power BI and viewed in Data View. It contains 32 rows, one per menu item, with four columns: menu_item_id, item_name, category, and price. Two DAX measures were created directly on this table: Monthly Revenue and Revenue by Month.

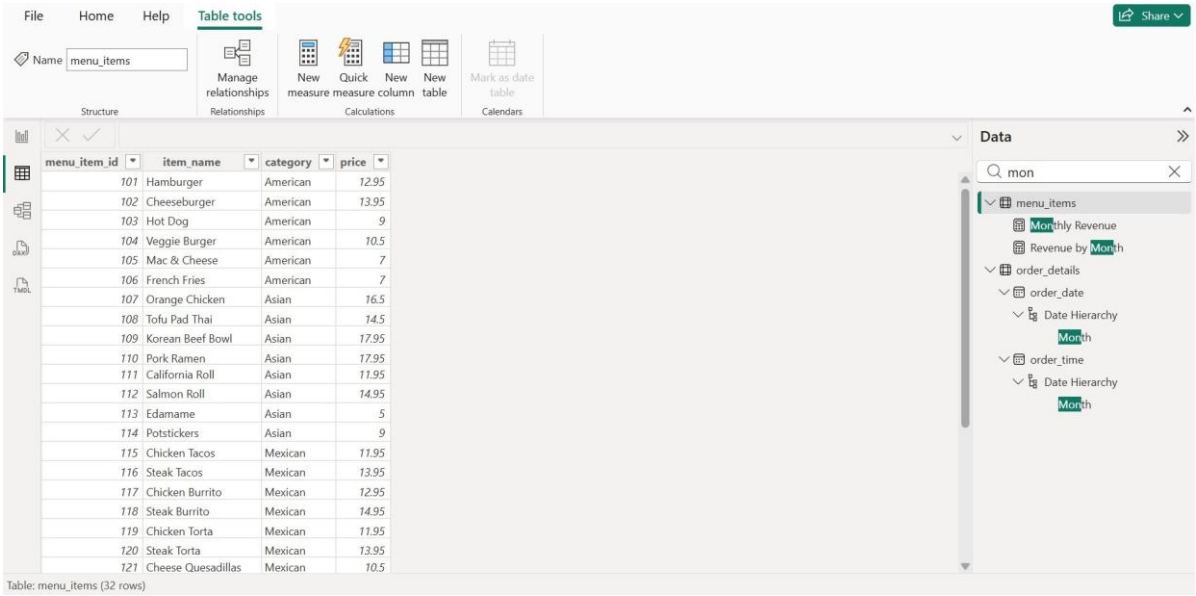


Figure 1 — Data View: menu_items table (32 rows) with DAX measures visible in the Data pane

1.2 Data View — order_details Table with Calculated Columns

The order_details table contains 12,234 rows and 5,010 distinct order_time values. After loading, three calculated columns were added using DAX to enable time-based analysis: Hour of Day, Day of Week, and Day Number. These columns are visible directly in the table alongside the original fields.

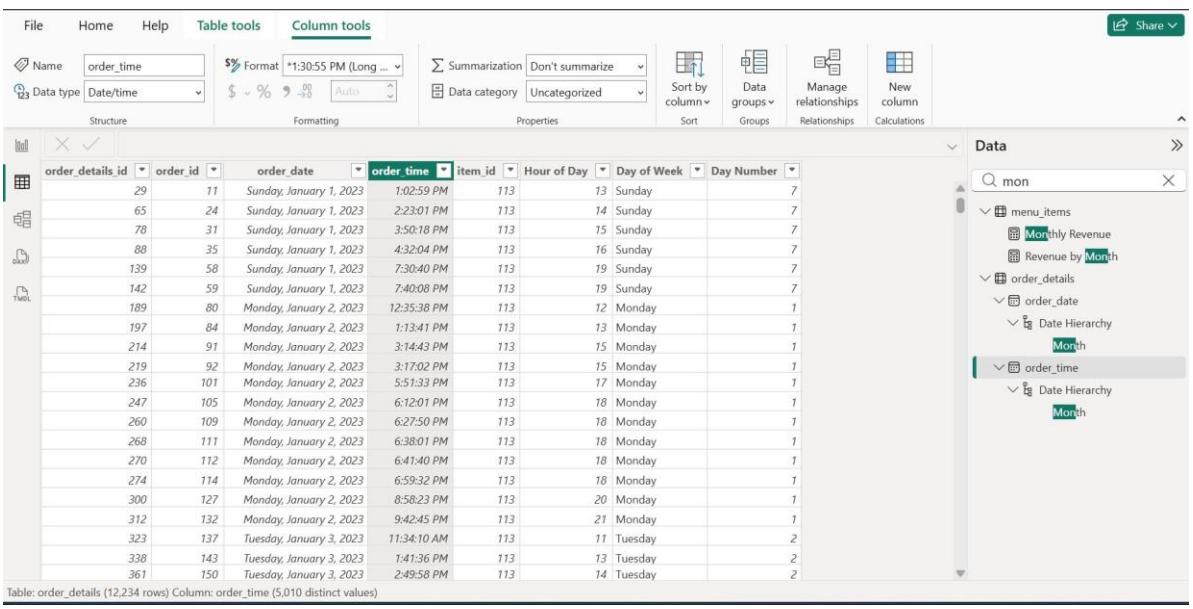


Figure 2 — Data View: order_details table with calculated columns Hour of Day, Day of Week, and Day Number

Calculated Columns Added to order_details

Column Name	DAX Formula Used	Purpose

Hour of Day	`HOUR( order_details[order_time] )`	Extracts hour 0–23 for hourly chart
Day of Week	`FORMAT( order_details[order_date], "dddd" )`	Extracts day name (Monday, Tuesday...)
Day Number	`WEEKDAY( order_details[order_date], 2 )`	1=Mon to 7=Sun — used to sort day axis

1.3 Model View — Relationship Between Tables

In Model View, a Many-to-One relationship was created by linking item_id in order_details to menu_item_id in menu_items. The relationship is shown as a line with 1 on the menu_items side and * (asterisk) on the order_details side. All DAX measures listed in the Data pane on the right were built on top of this relationship.

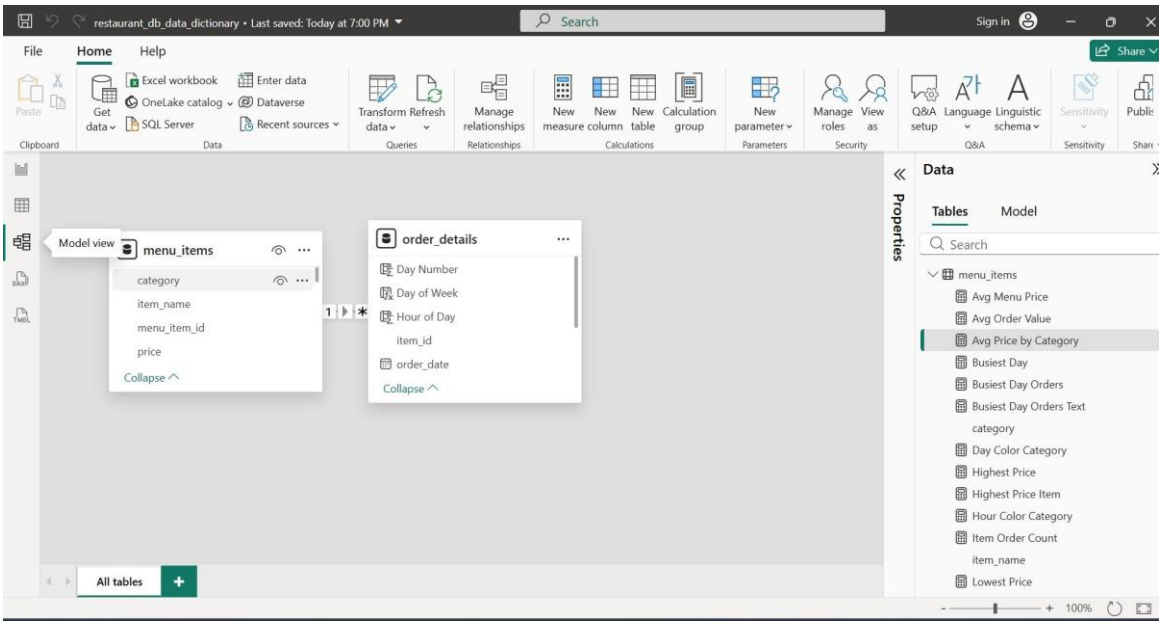


Figure 3 — Model View: Many-to-One relationship between menu_items (1) and order_details (*)

Setting	Value	Why
From table	order_details — item_id	Many orders reference each menu item
To table	menu_items — menu_item_id	Each item has exactly one record
Cardinality	Many to One (*:1)	Multiple orders per one menu item
Cross filter	Single	Filters flow from menu_items into orders

2.1 What Tools Were Used to Build KPIs

All KPI cards and chart values in this dashboard were built using DAX (Data Analysis Expressions) — the formula language built into Power BI. Each measure was created by clicking New Measure on the Table tools ribbon and writing a formula in the formula bar. Measures calculate dynamically based on whatever filters or slicers are active on the page.

2.2 KPI Measures Reference

Each measure below appears as a KPI card on one of the three dashboard pages:

Measure Name	DAX Formula	Dashboard Page
Total Revenue	`SUMX( order_details, RELATED( menu_items[price] ) )`	Sales Overview
Total Orders	`DISTINCTCOUNT( order_details[order_id] )`	Sales Overview
Avg Order Value	`DIVIDE( [Total Revenue], [Total Orders], 0 )`	Sales Overview
Total Menu Items	`COUNTROWS( menu_items )`	Sales Overview
Monthly Revenue	`CALCULATE( [Total Revenue], DATESMTD( order_details[order_date] ) )`	Sales Overview
Revenue by Month	`CALCULATE( [Total Revenue], ALLEXCEPT( order_details, order_details[order_date] ) )`	Sales Overview
Revenue by Cat %	`DIVIDE( [Total Revenue], CALCULATE( [Total Revenue], ALL( menu_items[category] ) ), 0 )`	Sales Overview
Item Order Count	`COUNTROWS( order_details )`	Menu Performance
Avg Menu Price	`AVERAGE( menu_items[price] )`	Menu Performance

Highest Price	`MAX( menu_items[price] )`	Menu Performance
Lowest Price	`MIN( menu_items[price] )`	Menu Performance
Avg Price by Category	`CALCULATE( AVERAGE( menu_items[price] ), ALLEXCEPT( menu_items, menu_items[category] ) )`	Menu Performance
Busiest Day	`TOPN on Day of Week by COUNTROWS DESC`	Time Analysis
Slowest Day	`TOPN on Day of Week by COUNTROWS ASC`	Time Analysis
Lunch Rush Orders	`CALCULATE( COUNTROWS( order_details ), [Hour of Day] >= 12, [Hour of Day] < 13 )`	Time Analysis

SECTION 3 — DATA CLEANING IN POWER QUERY

3.1 menu_items — Cleaning Steps

Column	Before Cleaning	After Cleaning
menu_item_id	Auto-detected as decimal	Set to Whole Number
item_name	Text — possible spaces	Trimmed + Text type
category	Text — possible spaces	Trimmed + Text type
price	Text or decimal	Set to Decimal Number

3.2 order_details — Cleaning Steps

Column	Before Cleaning	After Cleaning
order_details_id	Decimal	Whole Number
order_id	Decimal	Whole Number
order_date	Text / General	Date type — for all time analysis
order_time	Date/Time	Time type — for hourly analysis

item_id	Decimal	Whole Number — to match menu_item_id

3.3 Additional Steps

✓ Trim applied to item_name and category — removes hidden spaces that would break category grouping.

✓ No null, blank, or duplicate rows found in either table.

✓ Both tables merged with Left Outer Join: item_id (order_details) → menu_item_id (menu_items).

✓ Three calculated columns added to order_details: Hour of Day, Day of Week, Day Number.

✓ Relationship confirmed in Model View — Many-to-One, Single cross-filter direction.

SECTION 4 — DASHBOARD VISUALIZATIONS & FINDINGS

4.1 Sales Overview

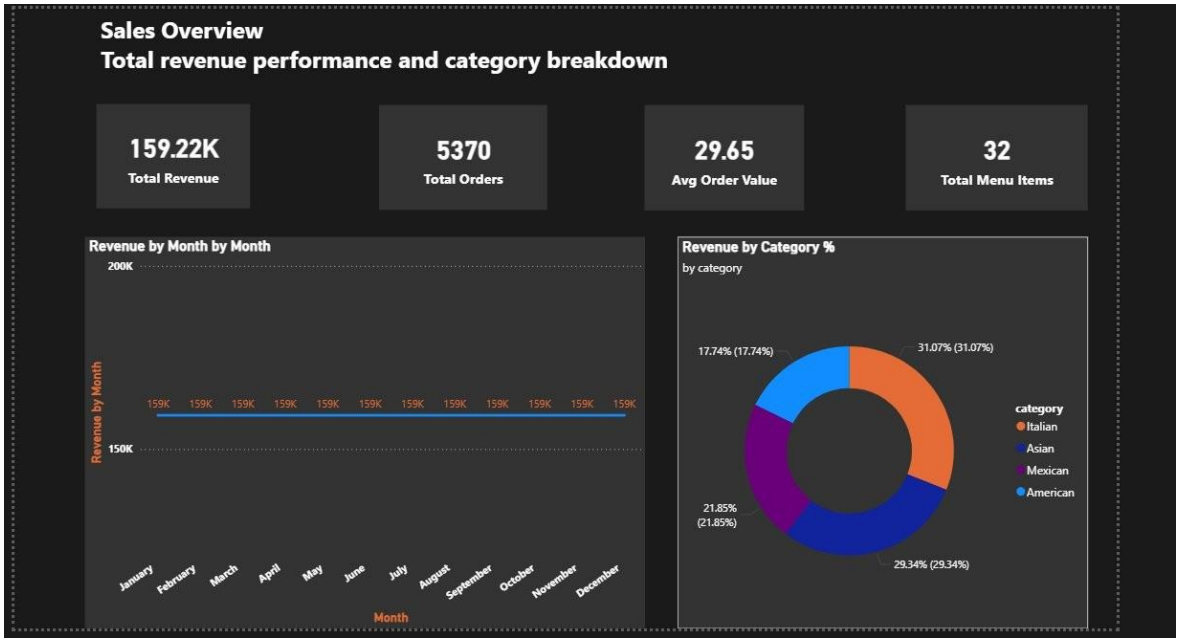


Figure 4 — Sales Overview: KPI cards, monthly trend, and revenue by category donut chart

KPI	Value	DAX Measure Used
Total Revenue	159.22K	SUMX(order_details, RELATED(menu_items[price]))
Total Orders	5,370	DISTINCTCOUNT(order_details[order_id])
Avg Order Value	29.65	DIVIDE([Total Revenue], [Total Orders], 0)

Menu Items	32	COUNTROWS(menu_items)

- Italian cuisine leads revenue at 31.07%, followed by Asian (29.34%), Mexican (21.85%), American (17.74%).
- Monthly revenue is flat at 159K every month — consistent demand throughout the year.
- Revenue by Category % measure used CALCULATE + ALL to compute each category's share of total.

4.2 Menu Performance

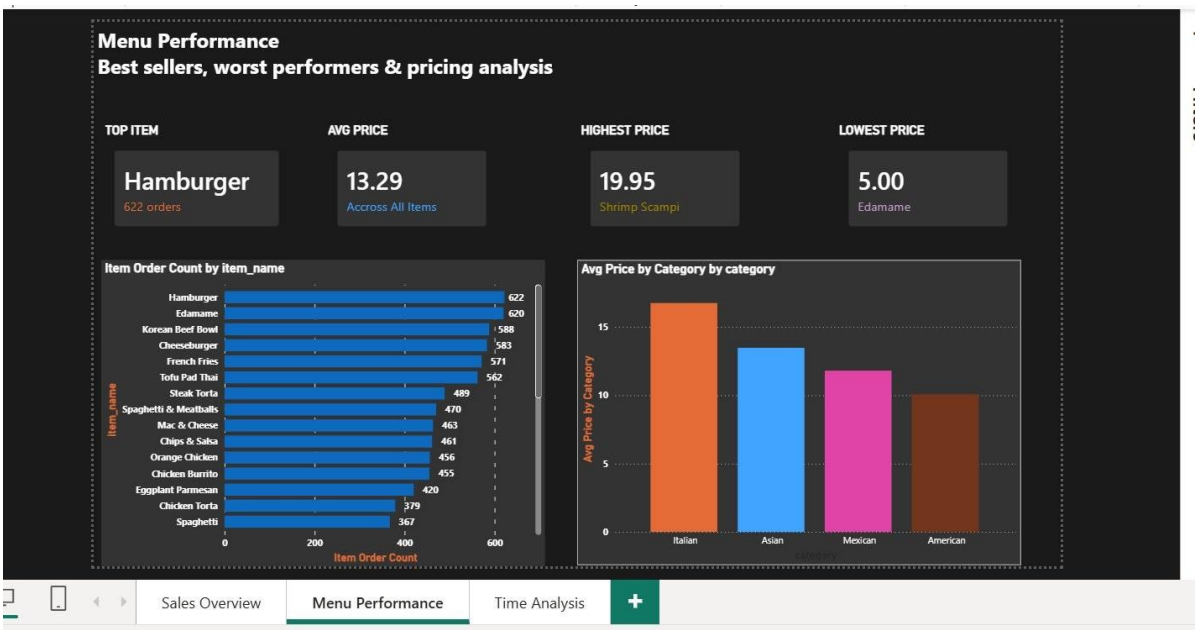


Figure 5 — Menu Performance: best sellers bar chart and avg price by category

KPI	Value	DAX Measure Used
Top Item	Hamburger (622 orders)	TOPN + COUNTROWS(order_details)
Avg Price	$13.29	AVERAGE(menu_items[price])
Highest Price	$19.95 — Shrimp Scampi	MAX(menu_items[price])
Lowest Price	$5.00 — Edamame	MIN(menu_items[price])

- Hamburger leads at 622 orders. Edamame (620) close second despite being the cheapest item.
- Italian category has the highest average price (~$17) — premium tier.
- Item Order Count bar chart uses COUNTROWS(order_details) filtered per item via the relationship.

4.3 Time Analysis

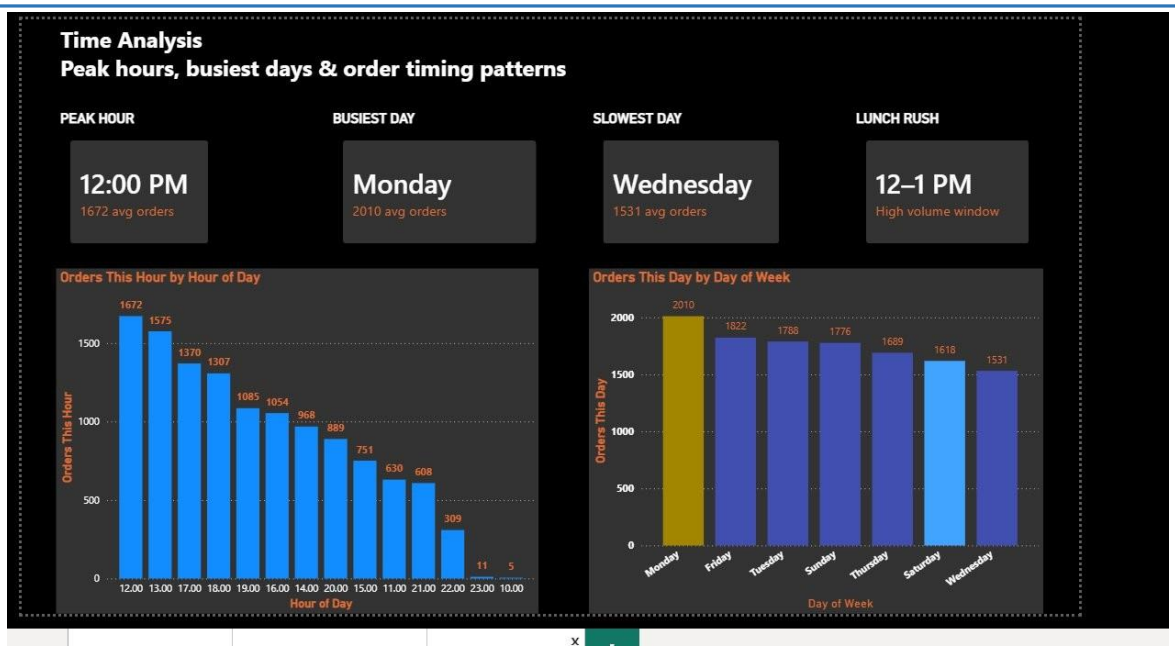


Figure 6 — Time Analysis: orders by hour of day and orders by day of week

KPI	Value	DAX / Column Used
Peak Hour	12:00 PM (1,672 orders)	Hour of Day calculated column + COUNTROWS
Busiest Day	Monday (2,010 orders)	Day of Week column + TOPN DESC
Slowest Day	Wednesday (1,531 orders)	Day of Week column + TOPN ASC
Lunch Rush	12–1 PM window	CALCULATE(COUNTROWS, [Hour of Day] >= 12 AND < 13)

Area	Key Finding	Action
Revenue	159K — flat monthly trend	Grow avg order value through upselling
Top Category	Italian leads at 31%	Promote Italian high-margin items

Best Seller	Hamburger — 622 orders	Feature in combo deals

Peak Time	12 PM — 1,672 orders/hour	Maximize lunch staffing & prep
Slowest Day	Wednesday — 1,531 orders	Run Wednesday-only promotions

- 12 PM is the single busiest hour — lunch dominates. Orders decline steadily through the afternoon.
- Monday is the busiest day (2,010 orders) — unusual pattern, worth investigating for a recurring event.
- Day of Week axis was sorted using Day Number column (1=Mon to 7=Sun) to avoid alphabetical ordering.

Summary

This project used Power Query for data cleaning and type correction, DAX for all KPI calculations, calculated columns for time intelligence, and Power BI visuals to present three fully interactive dashboard pages. The data model links both tables through a verified Many-to-One relationship enabling accurate cross-table analysis.